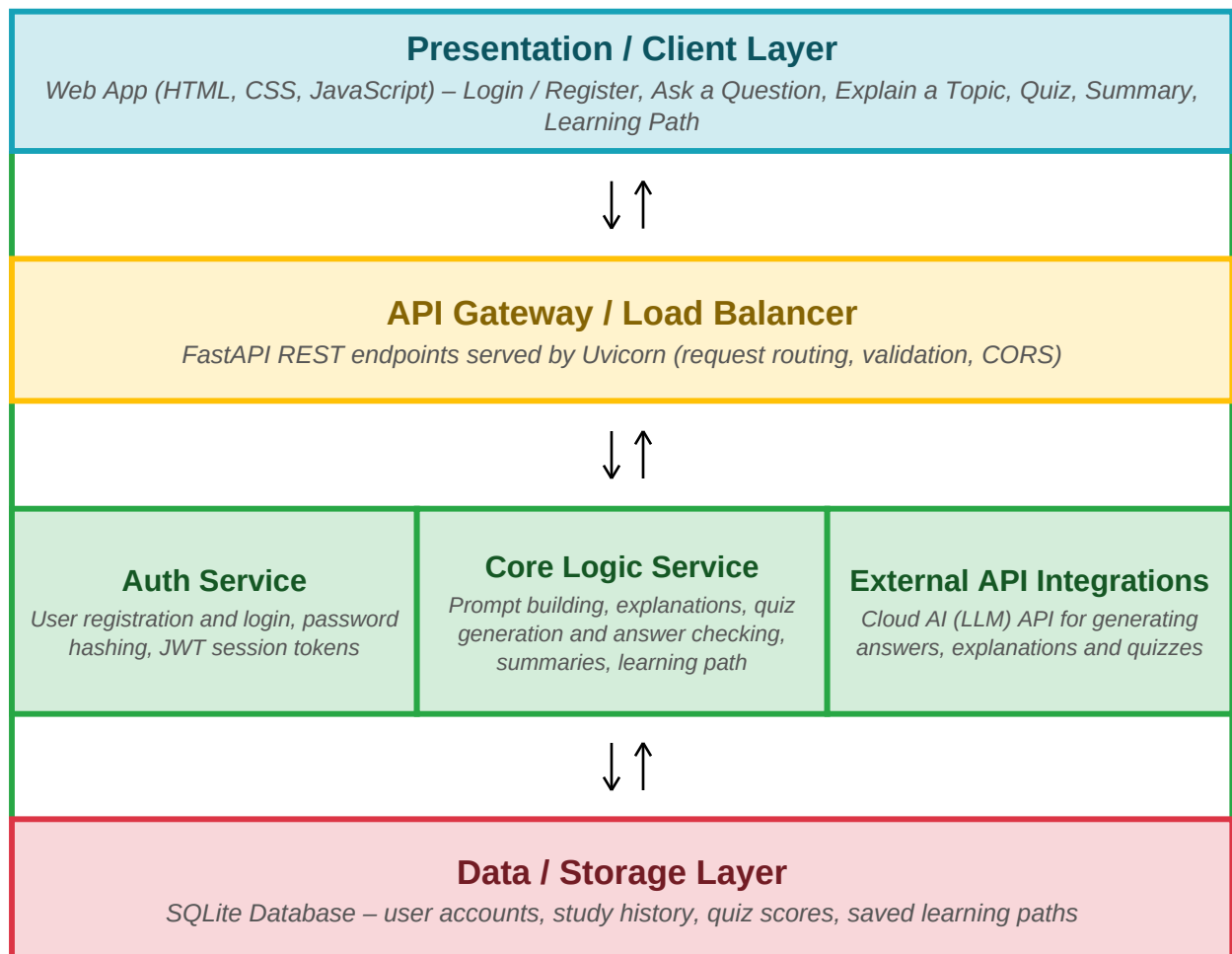


Solution Architecture

Date	29 September 2026
Team ID	UNM1659GenieBirds
Project Name	EduGenie – AI-Powered Learning Assistant
Maximum Marks	5 Marks

Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	The web interface students use. It handles login and registration and provides pages to ask a question, get a topic explanation, take a quiz and check answers, view a summary and see the recommended learning path. It sends requests to the backend and shows the results.	HTML, CSS, JavaScript (Fetch API)
API Gateway	The single entry point for all client requests. It routes each request to the correct service, validates input, applies CORS rules and returns JSON responses.	FastAPI, Uvicorn, Pydantic
Microservices (Auth, Core Logic, External API)	Auth Service registers users, verifies logins and issues tokens. Core Logic Service builds prompts, generates quizzes, checks answers, creates summaries and recommends the learning path. External API Integration sends prompts to the cloud AI model and returns its output.	Python, JWT, password hashing, Cloud AI (LLM) API
Database	Stores user accounts, question and explanation history, quiz scores and learning path progress. The data is small, so a lightweight file-based database is enough.	SQLite